Theoretical Background:

This project aims to design a water level indicator circuit that utilizes a NOT gate (7404 IC) to display the water level in a container using LEDs. The project utilizes the conductive properties of water. Bare wires act as water level sensors positioned at different heights within the container. As the water level rises, it completes the circuit between the ground (negative terminal) and the corresponding sensor wire. When a sensor wire (input) is dry (no connection to ground), the NOT gate receives a high logic signal. Since it's an inverter, the output becomes low. This low output, through the transistor and resistor, keeps the corresponding LED off. Conversely, when the water level rises and touches a sensor wire, the circuit is completed. The NOT gate input becomes low (grounded), causing its output to go high. This high output, amplified by the transistor, allows current to flow through the LED, illuminating it.

Here's an overview of the project:

7404 IC: This integrated circuit contains six independents NOT gates. Each gate inverts the input logic level (high becomes low and vice versa).

BC547 Transistor (NPN): This NPN transistor acts as a current amplifier. The low output from the NOT gate controls the base current, allowing a higher current to flow through the collector-emitter path, illuminating the LED.

Breadboard: This solderless prototyping platform allows for easy component connection without permanent soldering. Wires are inserted into the breadboard's holes to create temporary circuits.

7805 Voltage Regulator: This component ensures a stable 5V output from the 9V battery. LEDs typically operate at lower voltages, and the regulator protects them from damage caused by the higher voltage.

LED Indicator: Light-emitting diodes provide visual cues about the water level. Different LEDs illuminate based on the water reaching specific sensor points in the container.

Objectives:

1. Design a water level indicator circuit using a NOT gate.
2. Visually display the water level in a container (empty, low, half-full, full).
3. Utilize LEDs to provide clear visual representation about the water level.

Components Required:

Serial no.	Components Name	Specification	Quantity
1.	NPN Transistor	BC547	4
2.	Voltage Regulator	7805	1
3.	Resistor	10k Ω	4
4.	Resistor	220 Ω	4
5.	LED	5mm	5
6.	NOT Gate	7404	1
7.	Breadboard	-	1
8.	Battery	9V	1
9.	Connecting wires	Male to Male	40
10.	Connecting wires	Male to Female	6
11.	Container	-	1

Circuit Diagram:

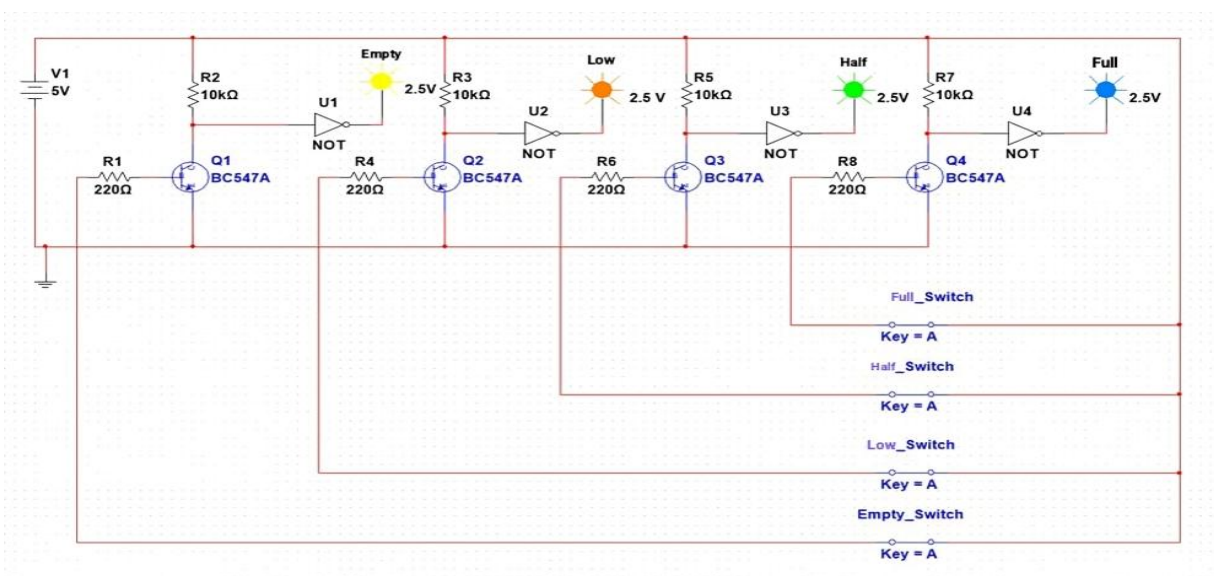


Fig: Circuit diagram of Water level Indicator Using NOT Gate

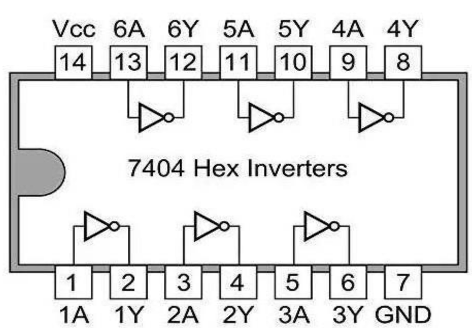


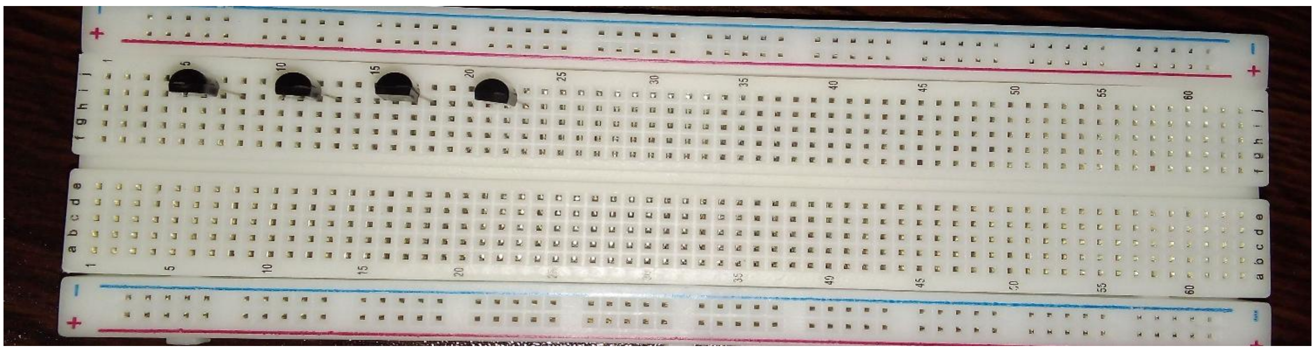
Fig: IC 7404 Pin Diagram

Hardware Implementation:

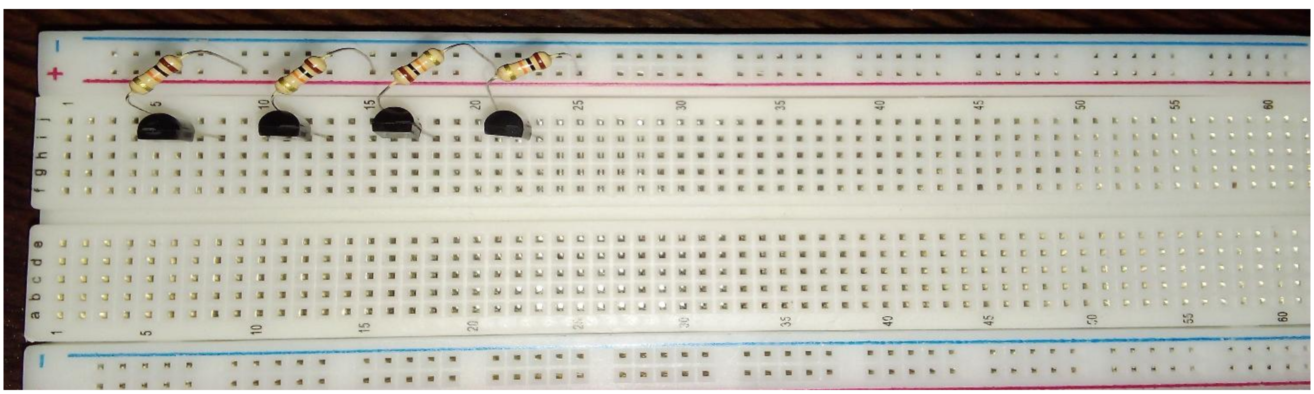
Drive Link:
https://drive.google.com/file/d/1VvY04woTt1_ATNp3qRwWP2Vsuy4SgIW4/view?usp=drive_link

Procedure:

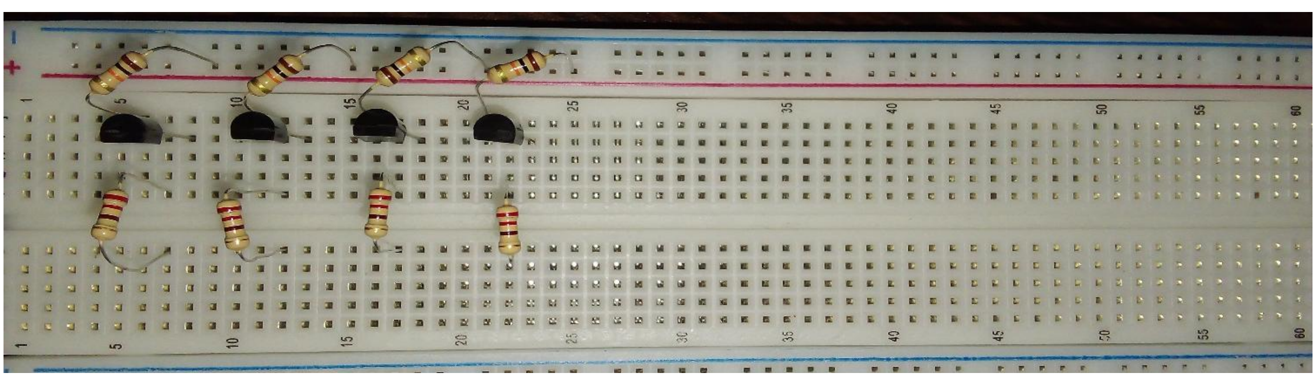
1. Insert 4 BC547 transistor on the breadboard. Here the left one is collector (C), the middle one is base (B), and the last one is the emitter (E).



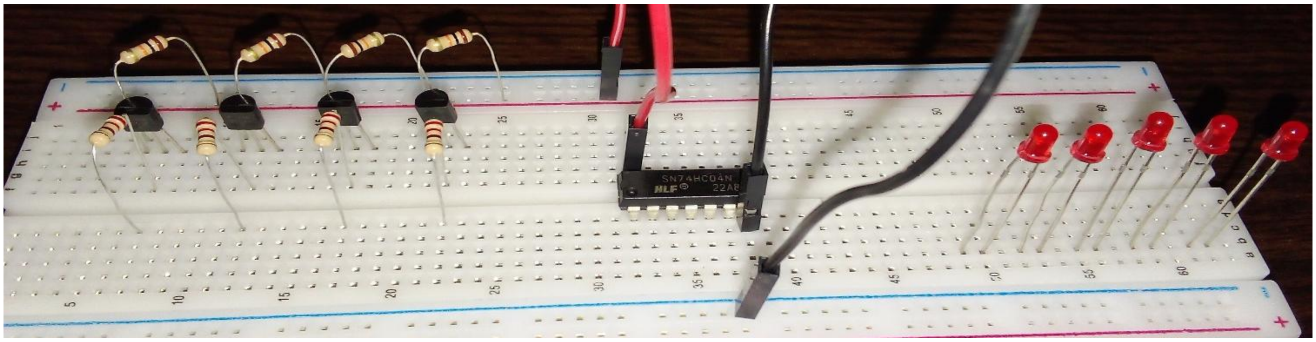
2. Connect 10kΩ resistors to each of the collector terminals of the transistor on the breadboard.



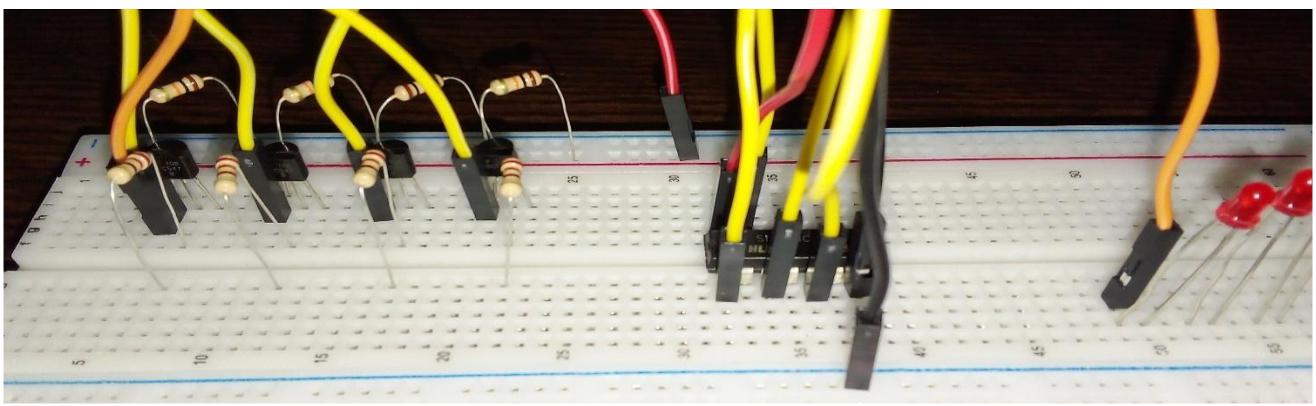
3. Connect 220Ω resistors to each of the base terminals of the transistor on the breadboard.



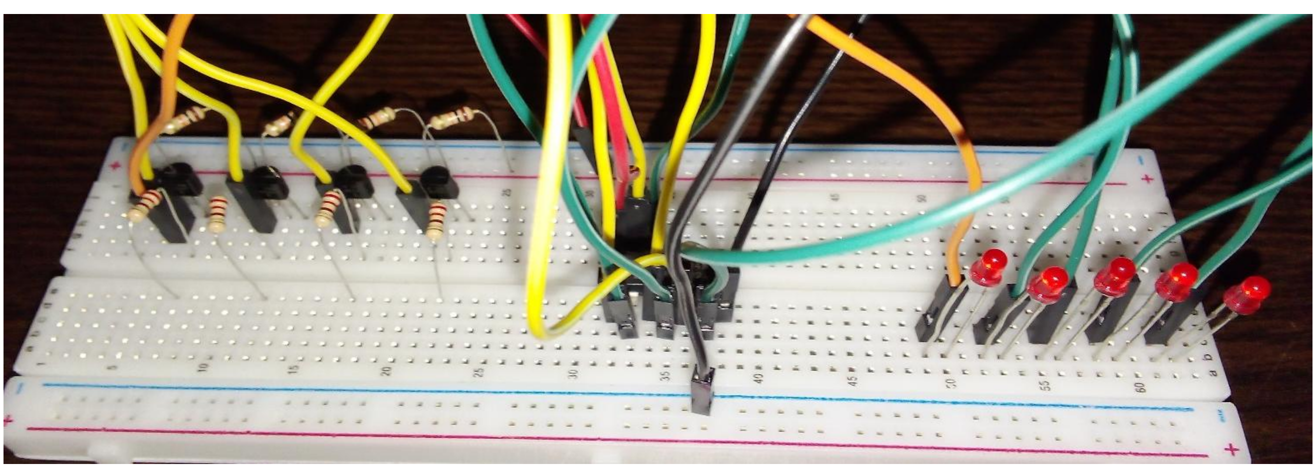
4. Connect a 7404 NOT Gate IC on the breadboard and connect the pin 7 (GND) to the ground and pin 14 (VCC) to the positive power supply voltage of the breadboard using two male to male connecting wires. Then insert 5 LED on the breadboard.



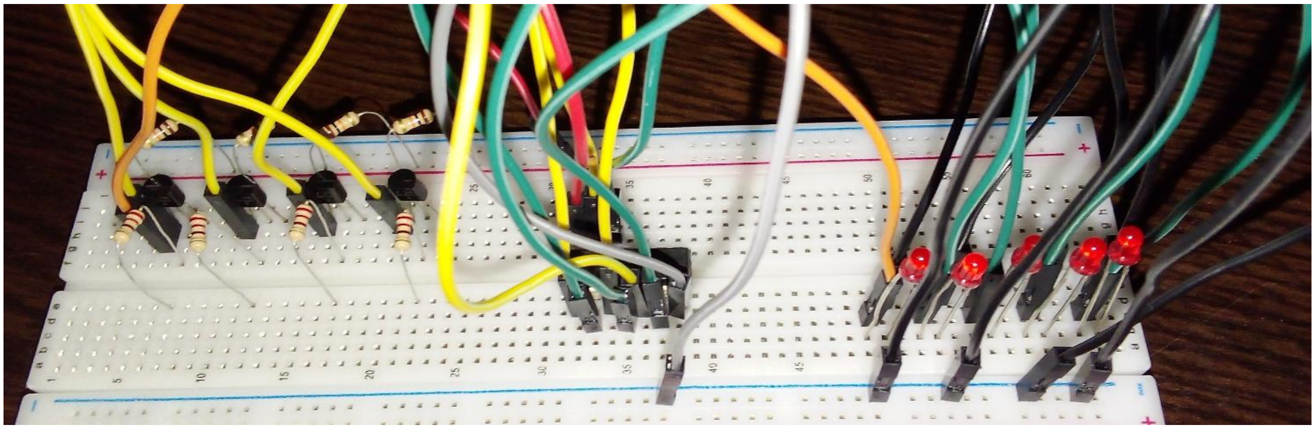
5. Connect 4 male-to-male connecting wires to the collector terminal of each transistor and take them to the input pins of the NOT Gate IC. Also, connect another wire to the emitter terminal of the 1st transistor and directly connect it to the anode terminal of the 1st LED so that when a voltage is applied to the circuit, it will indicate that the power is on.



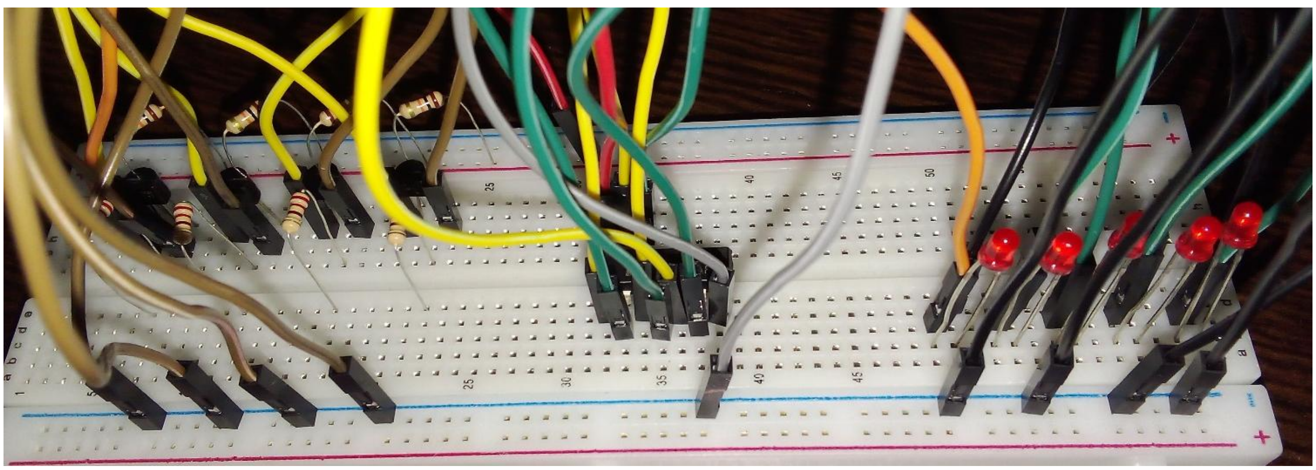
6. Now, from each output of the NOT Gate IC, insert wires and connect them to the anode terminals of the other 4 LEDs, which represent the empty, low, half, and full water levels respectively.



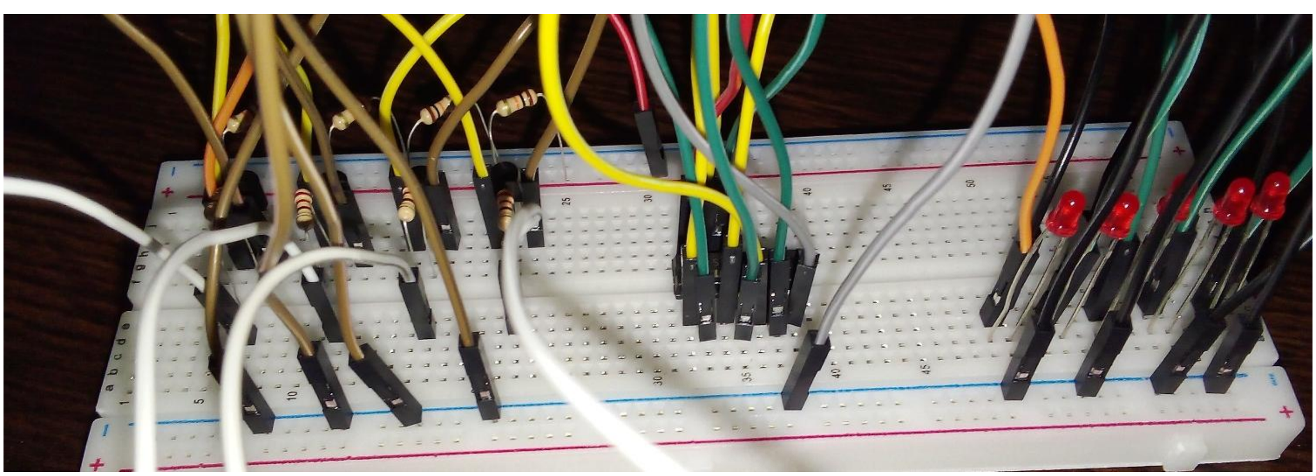
7. From each LED cathode terminal, insert a wire and connect it to the ground of the breadboard.



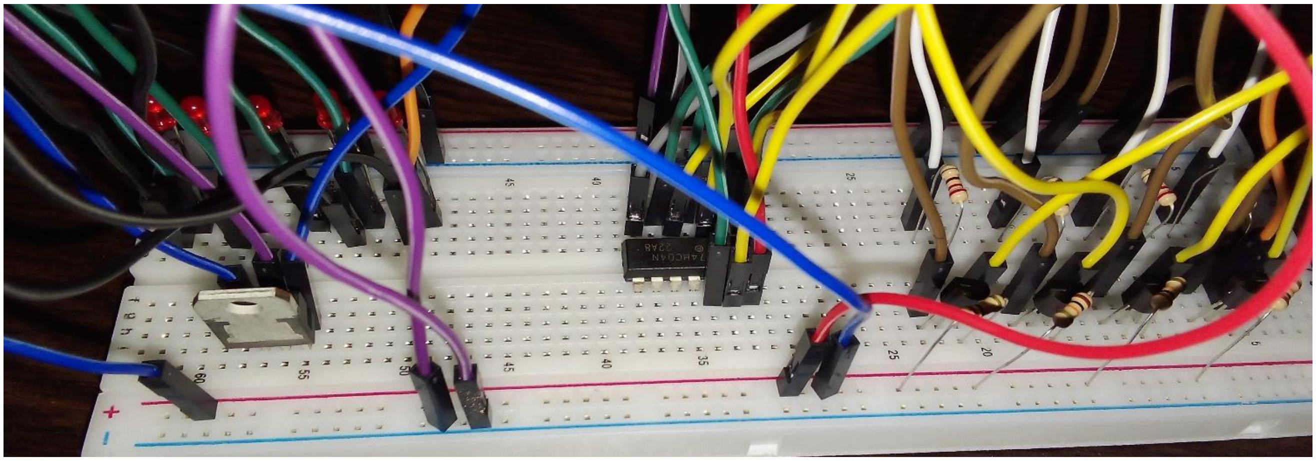
8. Connect all the emitter terminals of each transistor to the common ground of the breadboard.



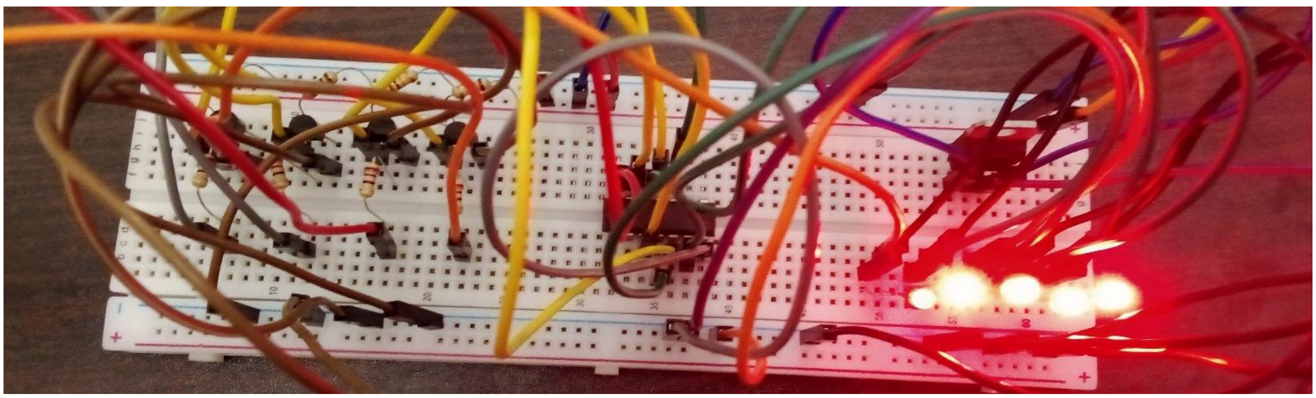
9. Connect 4 male to male connecting wires to the other side of 220Ω resistors.



10. Insert a voltage regulator to convert 9V to 5V on the breadboard.



11. To see the output, place the wires in the water container, connect a 9V battery to the circuit, and observe the water level indicated by the glowing LEDs.



Results:

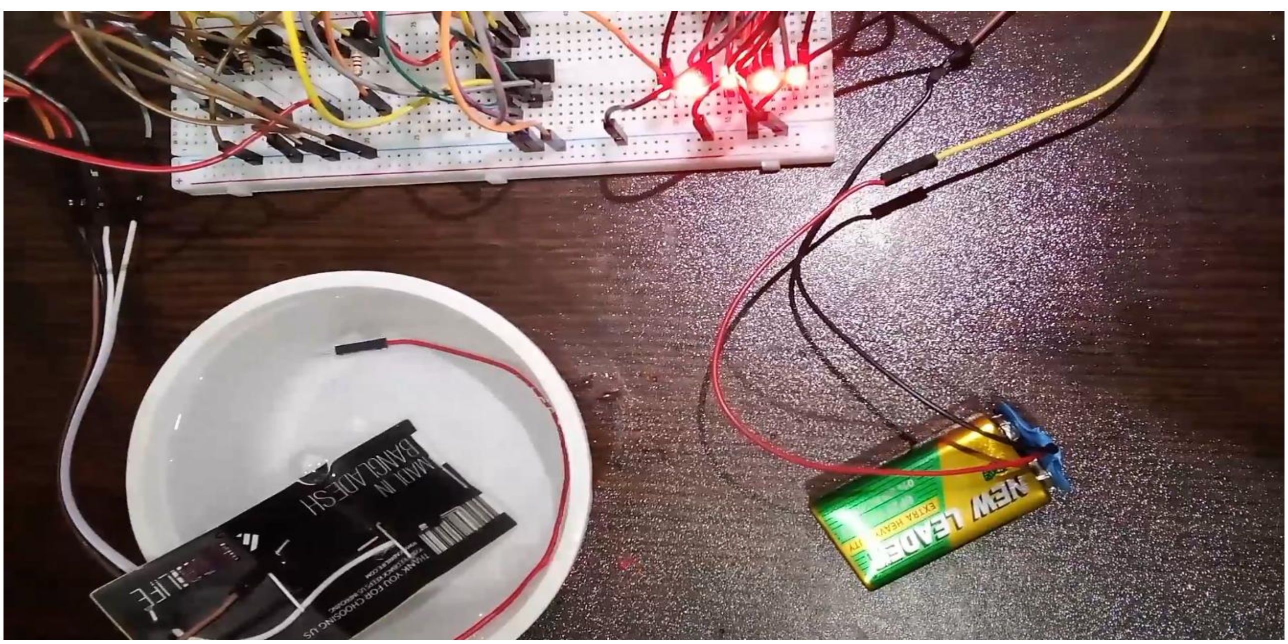


Image: Hardware Implementation

Truth Table for Water Level Indicator:

Water Level	Wire Input	Not Gate Output	Transistor State	LED State
System On	-	-	-	1st LED On
Empty	0	1	On	2nd LED On
Low	0	1	On	3rd LED On
Half	0	1	On	4th LED On
Full	0	1	On	5th LED On

Applications:

- Set up the indicator to trigger automated watering when plant pots reach a specific low water level.
- Place sensors in basements or crawlspaces to receive alerts for potential leaks before significant water damage occurs.
- Signal when a rain barrel for collecting rainwater is full, preventing overflow and potential damage to the collection system.
- Track water levels in a car wash bucket and receive a reminder to refill before running dry mid-wash.
- Track water levels in rivers, streams, or flood-prone areas to mitigate risks and ensure safety.
- Track proper water levels in swimming pools, notifying us when it's time to refill or adjust the water level to ensure enjoyment and safety.
- Track water levels in aquariums to ensure the water level remains optimal for the health of aquatic life.
- Track the home water tanks level efficiently and can prevent overflow or shortages.

Scope of Improvement:

- Integrate a buzzer to provide audible alerts or notifications alongside visual indicators.
- Upgrade to more accurate and reliable sensors for precise water level detection.
- Implement machine learning algorithms for optimized water level monitoring.
- Ensure the system is easily expandable and adaptable.
- Enable remote monitoring and control via wireless communication.
- Implement measures to protect against unauthorized access.
- Develop an intuitive interface for easy operation.
- Ensure seamless integration with other home systems.

- Optimize power consumption for cost-saving and environmental benefits.

Conclusion:

Our Water Level Indicator project utilizing NOT gates has successfully demonstrated a practical and effective solution for monitoring water levels in a container. By integrating basic electronic components, we have developed a system capable of accurately indicating different water levels, providing users with valuable information for efficient water management. As we continue to refine and improve upon our project, we emphasize the importance of simplicity, functionality, and reliability in engineering solutions for everyday challenges. This project serves as a testament to the versatility of electronic components and their potential to address various needs in both household and industrial settings.